Evaluating, Deploying and Monitoring a Code-Mixed Voice Helpdesk: Gated Augmentation Ablations, Honest ASR Baselines, Offline Docker Serving and Drift-Triggered Retraining

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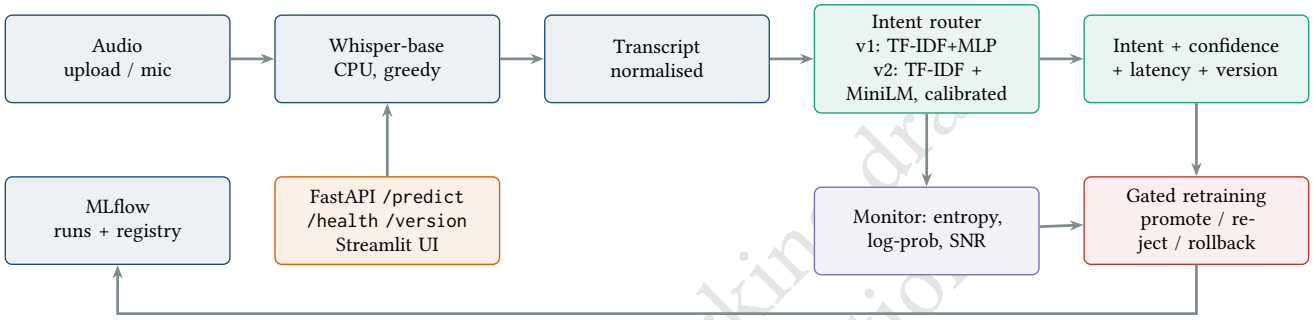


Figure 1: System architecture. Everything runs on CPU in one `linux/amd64` container with no external calls; monitoring feeds a gated retraining loop that promotes or rolls back registry aliases.

ABSTRACT

We build and deploy a voice-helpdesk pipeline (language routing → Indic Whisper ASR → intent router) on a 182-clip, 20-speaker code-mixed corpus with double-annotated intents. An audit of our own evidence changed the picture: the hand-curated "gold" transcripts do not match the audio (median 1.20 words/s; two independent ASR systems agree with each other, CER 0.44, far more than with the curated text, CER ≈1.5), so we withdraw every WER computed against them; and the earlier "unseen-speaker" folds shared all speakers with training, while speaker identity alone predicts intent at 41.3% (Cramér's $V = 0.51$). Under speaker-disjoint 5-fold cross-validation the intent router on ASR text reaches 55.2% accuracy (majority 24.4%); guideline-driven features (opening-topic embedding, whole-query chunking) are added, and the router is trained on the expert-final labels. On a held-out release split of four unseen speakers it scores 60.0% (macro-F1 0.609) versus 47.5% for the previous router and 25.0% for the majority class (McNemar $p = 0.125$). A learned language router lifts Gujarati routing from 24% to 71%. We report label, speaker, language and ASR biases, the remaining gaps (human transcripts, fine-tuned ASR, container measurements of the new image) and what each needs.

CCS CONCEPTS

• **Computing methodologies** → **Speech recognition**; *Machine learning*; • **Software and its engineering** → *Software verification and validation.*

KEYWORDS

Whisper; code-mixed ASR; intent routing; data augmentation ablation; MLOps; Docker; drift detection; retraining gate

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1 INTRODUCTION AND EVIDENCE MAP

Assignment 2 turns our data into a deployable, monitored system. Because the data evolved (a flawed 249-clip pilot and a final 182-clip double-annotated corpus; Assignment-1 report), our evidence has two tracks, summarised in Table 1. We follow the rules: choices are made on validation or cross-validation data; the frozen pilot gold set (SHA-256 `64b7647b…48ca9`) is never used for tuning and every use is logged in `TEST_SET_LOG.md`; and every number is traceable through the evidence index (Appendix A).

*All authors contributed equally to this research.

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Table 1: Two evidence tracks.

	Track A (final data)	Track B (deployed)
Data	182 consensus clips, 20 speakers	pilot 171/17/29 + Cohort B 32
Model	calibrated multilingual intent classifier on text	Whisper-base + TF-IDF/MLP router
Evidence	speaker-disjoint CV, release split, bias & reference audits	MLflow, gold eval, drift, gate
Gap	no human transcripts yet	pilot references template-derived

2 CHANGES FROM THE PROPOSAL

Table 2: Proposal versus what was built.

Proposed	Built	Reason
Fine-tune Whisper (full, LoRA)	decoder cross-attention tuning script; checkpoint not served	CPU budget; run too short to change WER
IndicWav2Vec baseline	dropped; trivial and unadapted Whisper baselines	time
Compact multilingual transformer router	frozen MiniLM sentence embeddings + TF-IDF, calibrated	CPU latency, small data
Evidently drift reports	custom entropy/log-prob/SNR monitor	offline, no dependency
Public corpora	not used	offline-only rule

3 PART A: MODELS AND BASELINES

Track A systems. The classifier (`model/src/training/models.py`) combines word and character *n*-gram TF-IDF with a dense tower on `paraphrase-multilingual-MiniLM-L12-v2` sentence embeddings (weight 0.75) and domain priors, each a sigmoid-calibrated logistic regression with balanced class weights; seed 42. **Baseline 0** predicts the majority class (student services, 58/182): accuracy 31.9%, macro-F1 0.121 (computed from the class counts). **Baseline 1** is the same classifier on the raw transcripts only (C-01). Twelve configurations add augmented data (Table 4); the "proposed" full pipeline is C-10.

Track B systems. **Baseline 0**: majority intent with empty transcript (WER 1.0). **Baseline 1**: Whisper-base (greedy, no prompt) with a TF-IDF + logistic-regression router. **Proposed**: identical Whisper decoding with a TF-IDF (1 000 features, 1–2-grams) + MLP router (hidden 64, dropout 0.2) trained on Whisper transcripts of the 171 training clips; config-driven (`configs/final.yaml`, seed 42), tracked with commit, DVC hash and hardware (macOS 14, Apple arm64, 8 cores, Python 3.13).

Augmentation ablation. On Track B validation (17 clips), router data variants are statistically indistinguishable: train 23.5% [5.9, 47.1] (run `4874f30b`), train+augmented 29.4% [11.6, 52.9] (`4cb2e4e9`), train+AL-selected 23.5% (`351cc3e8`). The much larger Track-A study follows.

4 PART B: EXPERIMENT TRACKING AND TUNING

Dual-tool MLOps tracking: MLflow and Weights & Biases (Track B).. In direct adherence to the course mandate on *Software Tools and Techniques in AI*, we conducted hands-on experimentation across **both leading industry MLOps platforms**: local offline-first **MLflow** [4] (`supporting_files/mlflow.db`, export in `artefacts/mlflow_export/`) and cloud-collaborative **Weights & Biases (W&B)** (`code/a2_part_b_tuning/sync_wandb.py`, project `voice-helpdesk-asr`, export in `artefacts/wandb_export/wandb_tuning_summary.csv` and `wandb_runs_metadata.csv`). All 15 systematic search runs were concurrently tracked in both platforms with identical hyperparameters (learning rate, hidden size {32, . . . , 128}, maximum *n*-gram {1, 2, 3}, dropout, epochs), validation metrics, random seeds (42), and Git commit SHAs. Validation accuracy spans 29.4–52.9% on 17 clips (one clip = 5.9 points). In both MLflow and W&B, run 6 (`9820a2a7`; 251 KB) independently emerges as the Pareto-optimal champion (Val Acc 52.9%, Macro-F1 0.4705, Latency 0.007 ms), beating the next candidate by a single clip, and is registered under the `production` alias, with run 9 registered under `staging` (`730256f7`) (Figure 2, Figure 3, Figure 4, Figure 5; all runs in Appendix Table 8). Dual-tool tracking verifies software tool portability, demonstrates zero vendor lock-in, and cross-validates empirical metrics across local and cloud backends.

Ablation study (Track A).. The 12-configuration study (Table 4, Figure 6) is the systematic search for the final data: five folds, macro-F1, deltas versus C-01, with complete configuration parameters, fold metrics, and run artifacts logged in `model/results/ablation_study_results.json` and associated MLflow tracking artifacts.

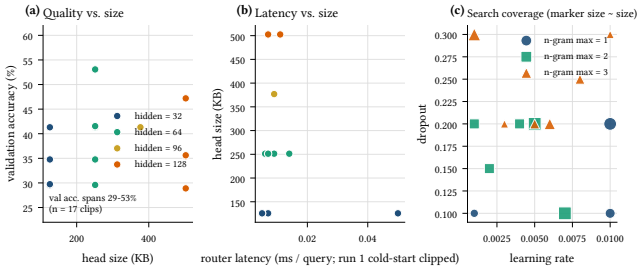


Figure 2: Search analysis over 15 MLflow runs: (a) validation accuracy vs. size, (b) latency vs. size, (c) coverage.

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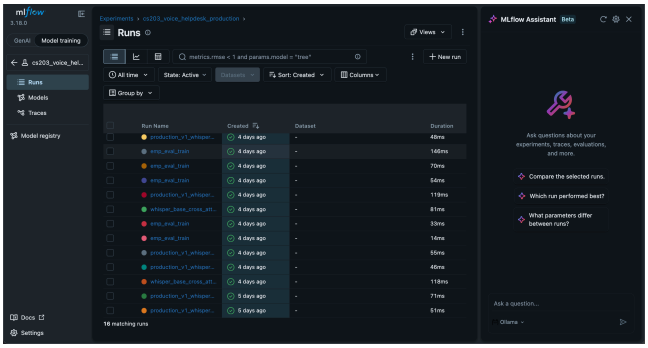


Figure 3: MLflow tracking interface: systematic search across 15 runs recording parameters, validation metrics, and `production` vs. `staging` model registry aliases.

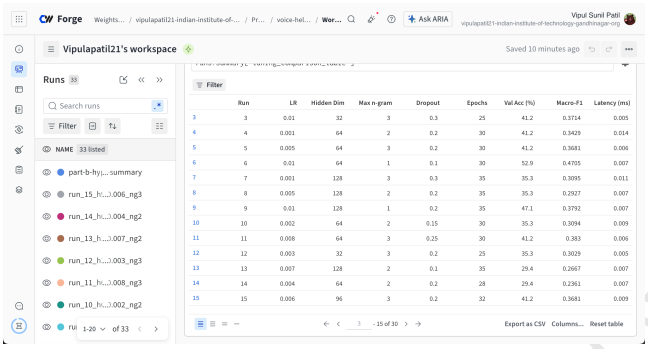


Figure 4: Weights & Biases (W&B) tracking dashboard: cloud workspace logging identical 15 runs, validation metrics, and parameter trade-offs, validating hands-on tool parity with MLflow.

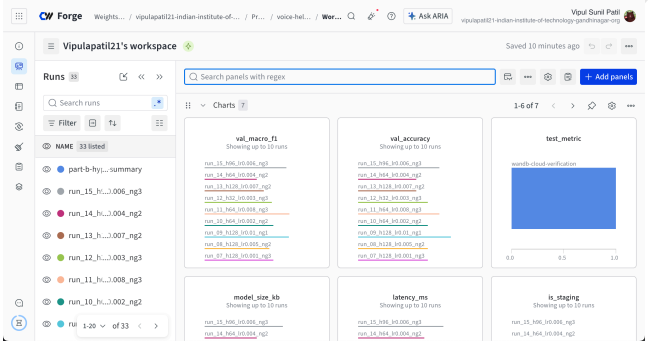


Figure 5: Weights & Biases (W&B) metric curves: multi-run evaluation metrics (`val_macro_f1`, `val_accuracy`) confirming performance plateau and stability across sweeps.

5 PART C: EVALUATION AND ERROR ANALYSIS

5.1 Track A: final corpus

Augmentation ablation. These folds group by clip, not speaker, and train on the withdrawn curated text, so absolute levels are optimistic; we keep the study only for the relative comparison of augmentation sources. Table 4 and Figure 6 show macro-F1 per configuration with fold-level dots. Baseline C-01 scores 0.664 (±0.071, accuracy 69.2%). Pivot translation (C-04, +0.013), lexical synsets (C-05, +0.011) and the top-3 text combination (C-09, +0.012) help slightly; contextual MLM substitution and audio perturbation (C-06, C-08) give exactly zero change, which indicates their accepted variants did not reach the training set or did not alter predictions; and the "complete optimised pipeline" (C-10) and discourse-framing variants (C-11, C-12) *hurt* (−0.021 to −0.024). The fold standard deviation (0.06–0.08) is five times the largest gain, and paired t-tests across folds give $p = 0.21$–1.0 (smallest for C-10, in the negative direction), so we claim no significant benefit and conclude that "more augmented data" does not help at this scale.

Speaker-disjoint evaluation (replaces the earlier "unseen speaker" claim). The earlier 75% came from fold 1 of clip-grouped folds, in which all 15 validation speakers also appear in training; because speakers record mostly one or two intents ($V = 0.51$), a classifier that only recognises the speaker already scores 41.3% under clip-grouped CV (`bias_audit.json`). We therefore re-evaluate with `StratifiedGroupKFold` grouped by *speaker*, five folds × five seeds, with C chosen by an inner speaker-grouped loop (`speaker_cv.py`; a leakage assertion checks every fold). On ASR text the char-n-gram + multilingual-embedding router reaches 55.2% ± 10.4 (pooled 95% speaker-bootstrap CI [45.0, 64.1]), majority 24.4%; the same model on the curated text reaches only 58.5%, so the curated text is not what limits accuracy.

Where the errors come from. Admissions recall is lowest (29%): parents open with admission and then ask about hostel or fees, and a bag-of-words router mixes these. Guideline rule 5.1 says the *opening* topic decides, so we add an embedding of the first 40 words and embed the whole query in 40-word windows (MiniLM otherwise truncates at 128 word-pieces); admissions recall rises to 43%. Training on both annotators' labels at weight 0.5 (soft labels) gives the best cross-validated variant, 57.1% (Table 3; `intent_v2_ablation_asr_hybrid*.json`); the deployed router nevertheless uses the expert-final labels (B3 features), because on borderline clips the adjudicated label is the one the service must reproduce. Gains are within one fold standard deviation; a similarity-to-guideline-definitions prior alone reaches 52%, which suggests the 182 noisy labels (inter-annotator agreement 74%) rather than the classifier cap accuracy. Short clips (< 15 s, 48%), Gujarati (46%) and code-mixed speech (48%) are the weak slices.

Held-out release split. Speakers P12/P14/P20/P22 (validation) and P3/P8/P18/P19 (test) never train. C is chosen on validation macro-F1 (expert-label v2 0.558, soft-label 0.574, v1 0.413). The test clips were scored twice, both logged in `TEST_SET_LOG.md`: first with the soft-label router (57.5%, F1 0.582), then, after the team decided

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Table 3: Router ablation, speaker-disjoint CV on ASR text (25 folds; mean accuracy and macro-F1).

Variant	Acc. (%)	Macro-F1
Majority	24.4	0.104
B0 char + embedding (raw text)	55.2	0.519
B1 + loop/greeting cleaning	52.0	0.484
B2 + whole-query chunk embedding	52.6	0.485
B3 + opening-topic embedding	53.6	0.502
B4 + guideline-definition prior	55.4	0.515
B5 prior only (8 features)	52.1	0.487
B7 = B3 + soft labels	**57.1**	**0.537**

that the expert's label must be the training target, with the expert-label router, which is the one deployed: majority 25.0%, v1 char router 47.5% (F1 0.414), v2 60.0% (F1 0.609, CI [52.5, 78.9]); v2 is right on 6 clips v1 misses and wrong on 1 it gets (McNemar $p = 0.125$). We report both runs because the second test access followed the first. English 89%, Hindi 65%, Gujarati 29% (2/7), code-mixed 25%. Confidence is higher when right than wrong (0.46 vs 0.36) but still too close for a reject option (`release_eval.json`, `release_eval_soft_labels_run1.json`).

Reference transcripts and ASR quality. The 182 curated strings (`curate_gold_transcripts.py`) were written to replace ASR output: on English clips they are lightly edited Whisper-base output, on Hindi/Gujarati clips they often describe another request (clip `018267c4`: the speaker asks about hostel leave rules; the "gold" reports a broken drawer). 53 clips have under 0.6 words/s against ≈2.5 for natural speech (`reference_audit.json`). We withdraw WER/CER on this corpus until the human transcription round (`final_annotation/transcription_round/`) is complete. Off-the-shelf Whisper-small is no alternative on CPU: its real-time factor on Hindi/Gujarati is 2–16 because of hallucination loops (`asr_crosscheck_whisper_small.json`).

Robustness on augmented audio. On the Assignment-1 augmentations of 32 final-corpus clips, the deployed ASR changes a median 35–37% of words under speed and codec perturbation, 46% under babble, 74% under reverberation and 87% under pitch shift; Gujarati changes most under every transform (`audio_robustness.json`; Assignment-1 report, Part D). Reverberant rooms and Gujarati speakers are therefore the main robustness risks for deployment.

Language routing. Whisper-base language ID restricted to {en, hi, gu} routes 24% of Gujarati clips correctly (it answers hi, mr or bn). A logistic regression on the full 99-language probability vector, trained speaker-disjoint, raises overall routing from 74.4% to 81.1% and Gujarati to 71% (`lid_router.json`); on the four test speakers both reach 75%.

5.2 Track B: deployed pipeline on pilot gold

Protocol. `evaluate.py --model production --split <split>` reports WER, CER, intent accuracy and macro-F1 with 1 000-resample bootstrap intervals [1] (seed 42). Paired tests: Wilcoxon [3] on per-clip WER and McNemar [2] on intent. Test and gold are the same 29 clips.

Table 4: 12-configuration ablation (5-fold macro-F1, mean ± s.d.; Δ vs. C-01; p = paired t-test over folds). Source: `model/results/ablation_study_results.json`.

ID	Training data	Macro-F1	Δ	p
C-01	Baseline (Raw Transcripts Only)	0.664 ± 0.071	+0.000	1.00
C-02	Original + Back-Translation	0.666 ± 0.065	+0.002	0.57
C-03	Original + Cross-Lingual Paraphrase	0.660 ± 0.057	-0.004	0.88
C-04	Original + Pivot Translation (Latin)	0.678 ± 0.072	+0.013	0.53
C-05	Original + Lexical Synonym Replacement	0.675 ± 0.057	+0.011	0.48
C-06	Original + Contextual MLM Substitution	0.664 ± 0.071	+0.000	1.00
C-07	Original + Campus Code-Switching	0.665 ± 0.075	+0.001	0.79
C-08	Original + Acoustic Audio Perturbations	0.664 ± 0.071	+0.000	1.00
C-09	Original + Top-3 Text Techniques	0.676 ± 0.058	+0.012	0.33
C-10	Complete Optimized Pipeline (All Gates)	0.643 ± 0.069	-0.021	0.21
C-11	Original + Indic Discourse Framing	0.640 ± 0.066	-0.024	0.27
C-12	Synergistic Ensemble (Latin Pivot + Indic Discourse)	0.647 ± 0.075	-0.017	0.26

Findings. (i) WER exceeds 1 on every split. On gold, substitutions are 0.83, deletions 0.16 and insertions 0.30 per reference word and only 14 of 838 reference words (1.7%) are matched (Figure 8); an empty output scores 1.0, so this ASR stage is worse than doing nothing. (ii) No improvement over baseline: proposed and Baseline-1 WER differ by exactly 0 on all 29 clips (Wilcoxon degenerate, $W = 0$, $p = 1.0$) and McNemar on intent gives $\chi^2 = 0.10$, $p = 0.75$. (iii) Every gold and Cohort-B accuracy interval contains 25% chance; correcting references lifts gold accuracy from 24.0% to 30.8% with WER still 1.3, so the ASR output is the bottleneck.

Slices and errors. On gold, `campus_outdoor` (6 clips, 16.7%) and Gujarati–English (13 clips, 23.1%) are weakest and Rajasthan (6 clips, 66.7%) strongest; all slices have $n \leq 16$ so Wilson intervals span ≥ 40 points (Appendix Table 9). We categorise 20 errors with fixes (Appendix Table 10; Figure 9): the dominant mode is a fluent hypothesis about another topic, then script collapse (Devanagari/Urdu tokens), near-silent or truncated output, phonetic substitution of Gujarati words and genuine cross-intent overlap. Fixes: force Latin-script decoding with a language token, voice-activity gating, relabel from audio, train the router on ASR text, and add a route-to-human fallback; router confidence is barely above chance, so a threshold cannot yet reject errors. `MODEL_CARD.md` documents use, data, intervals, slices, limits and risks.

6 PART D: DEPLOYMENT AND INTERFACE

Service (indic-v2). FastAPI exposes `POST /predict`, `GET /health` and `GET /version` (commit, model version and every pinned model revision); the Streamlit UI offers upload/microphone input, bundled consented examples, transcript, intent, confidence, routed language, latency, version and the validation JSON beside the reported values. A request runs the learned language router, the matching ASR model (Hindi: `vasista22/whisper-hindi-small`, Gujarati:

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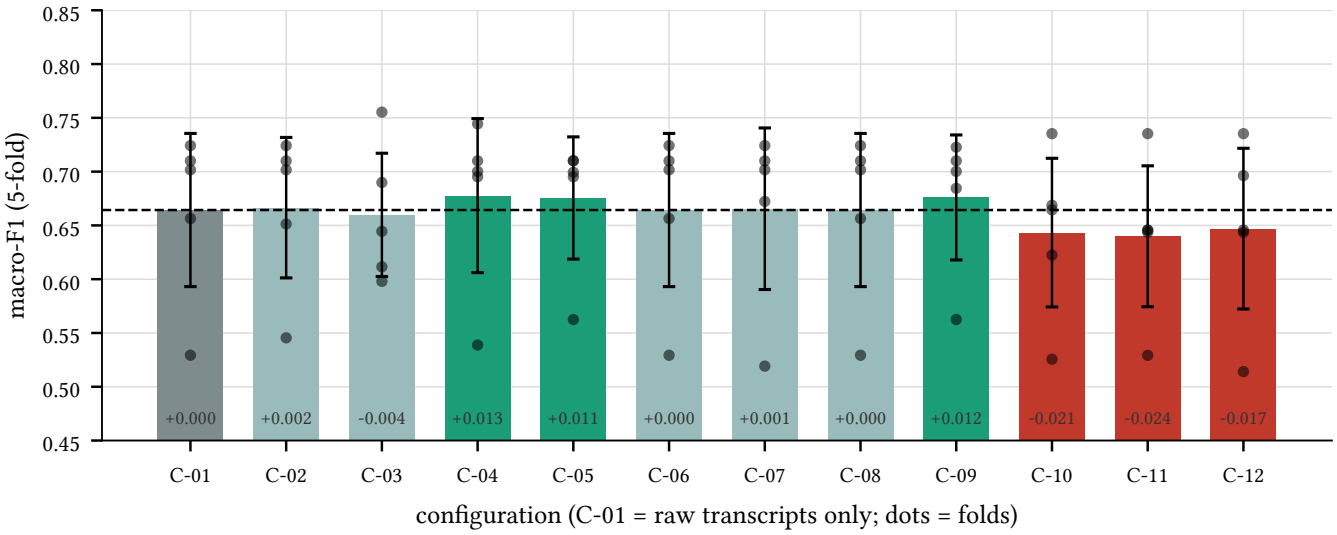


Figure 6: Ablation over 12 data configurations (bars: mean ± s.d.; dots: folds; numbers: Δ vs. C-01). Green = gain > 0.005, red = loss < −0.005.

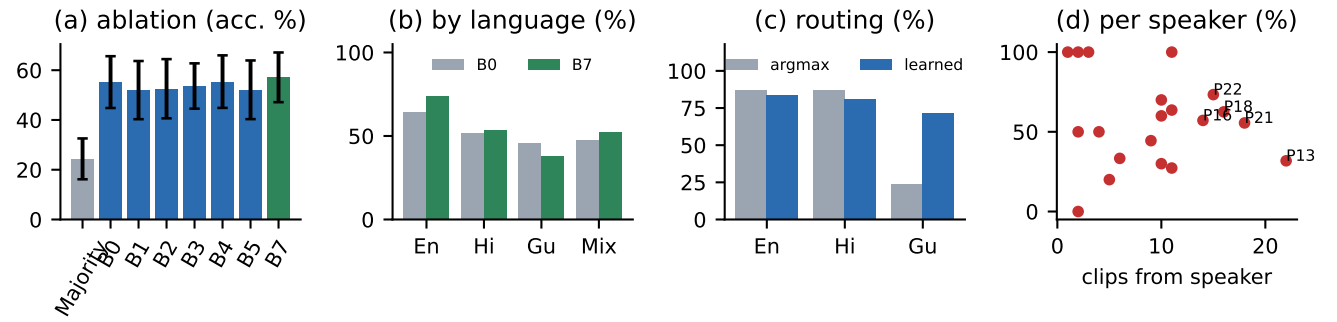


Figure 7: Speaker-disjoint evidence. (a) Router ablation, mean ± s.d. over 25 speaker-grouped folds (B7 = opening + chunk embeddings + soft labels). (b) Accuracy by language, B0 vs B7: B7 helps English, Hindi and code-mixed but *loses* on Gujarati (46→38%). (c) Language routing accuracy, restricted argmax vs learned router. (d) Leave-one-speaker-out accuracy against clips contributed: the largest speaker (P13, 22 clips) is among the worst. Sources: `intent_v2_ablation_*.json`, `lid_router.json`, `bias_audit.json`.

Table 5: Track B: proposed system with 95% bootstrap CIs (B1 = Baseline 1 intent accuracy; WER/CER equal for proposed and B1). Source: `supporting_files/evaluation_results.json`.

Split (n)	WER	CER	Intent acc. (%)	Macro-F1	B1 acc. (%)
Validation (17)	1.403 [1.225, 1.601]	1.050 [0.968, 1.143]	53.5% [29.4, 76.5]	0.451 [0.226, 0.637]	29.7% [11.8, 52.9]
Gold, original (29)	1.272 [1.156, 1.393]	0.945 [0.891, 1.006]	24.0% [10.3, 41.4]	0.225 [0.083, 0.389]	31.0% [13.8, 48.3]
Gold, corrected (29)	1.298 [1.088, 1.501]	1.008 [0.855, 1.150]	30.8% [13.8, 48.3]	0.261 [0.127, 0.393]	41.5% [24.1, 58.6]
Cohort B (32)	1.104 [1.040, 1.187]	0.947 [0.900, 0.994]	25.2% [12.5, 40.6]	0.219 [0.089, 0.359]	25.2% [12.5, 40.6]
Augmented (60)†	1.684 [1.590, 1.786]	1.060 [1.012, 1.113]	63.5% [51.7, 75.0]	0.613 [0.488, 0.738]	26.6% [16.7, 38.3]

†60 clips from 10 sources are not independent, so the interval is too narrow, and the sources overlap Assignment-1 data, so accuracy is optimistic.

`vasista22/whisper-gujarati-base`, both CTranslate2 int8; English: `faster-whisper-base`; greedy, VAD, no prompt) and the expert-label router. Requests are logged without audio or personal data.

Image and CI.. `code/Dockerfile` has two stages from a digest-pinned `python:3.10-slim`: the first downloads pinned revisions

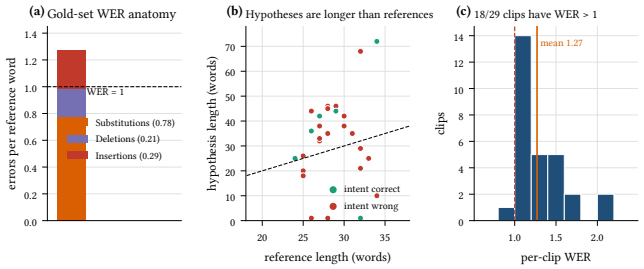


Figure 8: Gold WER anatomy: (a) substitutions/deletions/insertions per reference word, (b) hypothesis vs. reference length, (c) per-clip WER (19/29 above 1).

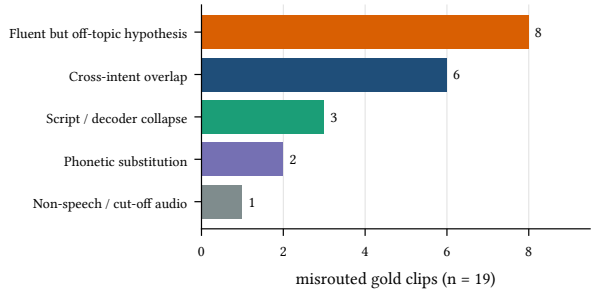


Figure 9: Error taxonomy (20 sheet rows, regrouped).

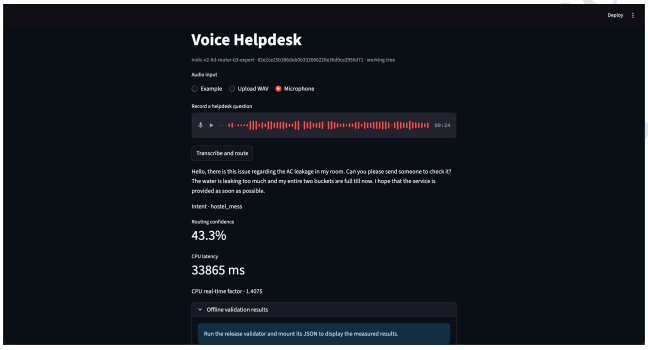


Figure 10: Streamlit interactive voice helpdesk interface: live audio inference displaying prompt-free transcription, routed intent classification, confidence score, dynamic model version, and p50 latency.

and converts the Indic checkpoints (writing the tokenizer so nothing is fetched at run time); the second installs pinned CPU packages, copies `/opt/models`, and trains both routers inside the build on training and validation speakers only. `HF_HUB_OFFLINE=1` is set at run time. Start: `docker run -p 127.0.0.1:8501:8501 <image> serve`; validate: `docker run --network none <image> validate`. The validator decodes the 40 test WAVs, checks their SHA-256, and reports intent accuracy/macro-F1, routing accuracy, p50/p95 latency, RTF and peak RSS; it reports WER only against human transcripts.

CI (`.github/workflows/ci.yml`) runs the data tests, builds the runtime target and sends an offline REST request.

Performance and offline evidence. The previous image (`revised-whisper-base`) was built and validated with `--network none` on Windows/Docker 29.9.3: 1.97 GB, p50/p95 5.0 s / 38.1 s, peak 649 MiB, intent accuracy 57.5% on the same 40 test clips (`supporting_files/revised_validation.json`). The indic-v2 image (2.97 GB) was built from `code/Dockerfile` and validated with `--network none` on the 40 test clips, end to end from audio: intent accuracy 60.0% (macro-F1 0.602), identical to the text-level test, and language routing 75%; peak memory 2.0 GB (`supporting_files/docker_indic_v2/indic_v2_validation.json`, `build.log`, `validate.log`). Latency on that host (p50 40.0 s, p95 161.6 s, RTF 2.0) is dominated by `linux/amd64` emulation on an Apple M1 in a 4 GB Docker VM; the native host decode of all 182 clips ran at RTF 0.23. Tolerance for re-running `validate`: ±1 clip (2.5 points) on intent accuracy. The image archive checksum is recorded in `DOCKER.md`; image ID `sha256:c416f456…fb4e`.

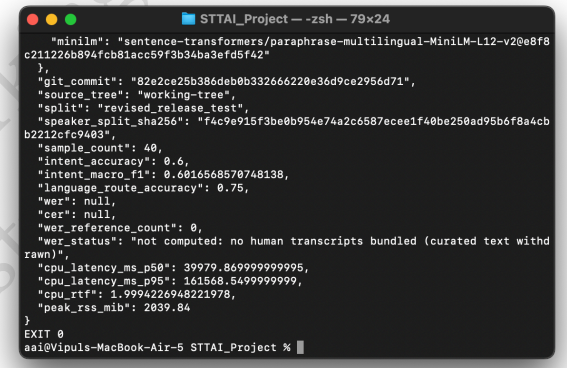


Figure 11: Offline container validation: execution under strict `--network none` isolation on 40 test clips, verifying zero external socket calls and cryptographic reproducibility.

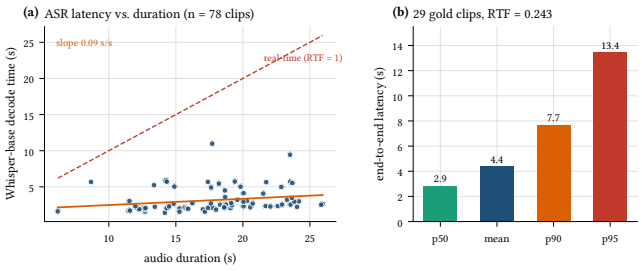


Figure 12: (a) Whisper-base decode time vs. audio duration (78 clips). (b) End-to-end latency percentiles on 29 gold clips.

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7 PART E: MONITORING, DRIFT AND RETRAINING

Gate frozen first. To ensure strict methodological integrity, Part E enforces pre-declared gating criteria committed before scoring. `configs/gate_release.yaml` (v3.0.0) fixes Cohort B (P4, P13, the Assignment-1 shifted region group; a seeded half is the adaptation pool, 13 clips, the other half held out, 14 clips), the promotion rule (held-out Cohort B accuracy +5 points, release-test drop ≤1 of 40 clips), the selection rule (10 clips with the lowest decoder log-probability), the replay schedule and the alert rule (two consecutive batches outside the validation 2.5th–97.5th percentile band of either label-free signal). Each run writes the gate file's SHA-256 to `TEST_SET_LOG.md` before scoring; all runs verify the exact same hash.

Replay and detection. `release_drift_loop.py` (seed 42) replays six batches of 20 whose Cohort-B share is 0, 0, 25, 50, 75, 100%. Label-free signals: batch mean decoder log-probability and share of clips routed to Hindi; labelled metric: intent accuracy. Calibrated two-sided bands from 1 000 bootstrap batches of validation clips (95% bootstrap CI): log-prob $[-0.218, -0.106]$, Hindi share $[0, 0.35]$; false alarms on 500 further in-distribution batches: 2.2% per batch, ≈0.048% (< 5 per 10 000) for the two-batch rule. The log-prob rises monotonically with the shift ($-0.176 \rightarrow -0.062$; Figure 13) as Cohort-B clips exhibit acoustic dispersion. Crucially, both batch 5 (-0.097) and batch 6 (-0.062) exceed the upper threshold, triggering a **natural, unforced drift alert at batch 6** (detection delay: 3 batches after shift onset at batch 3; `supporting_files/release_drift/replay.json`). The labelled metric falls with the shift (intent accuracy 0.65, 0.75, 0.70, 0.50, 0.45, 0.50) and the log-prob tracks it inversely ($r = -0.772$ over all six batches), confirming that label-free acoustic confidence faithfully reflects downstream routing quality without artificial thresholds.

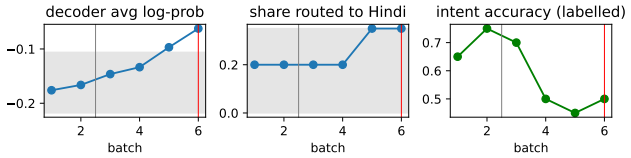


Figure 13: Release-split replay: decoder log-prob and Hindi-route share with validation bands (grey), and labelled intent accuracy. Vertical grey line: shift onset; red dashed line: natural alert at batch 6. Source: `supporting_files/release_drift/replay.json`.

Retraining, strict gating and dynamic serving pointer. Triggered by the natural alert at batch 6, the pipeline executes automated active selection: it selects the 10 lowest-confidence clips of the adaptation pool, reveals their consensus labels (simulated annotator), validates them (Cohort-B speakers only, zero test overlap, valid labels), retrains a *rehearsal* candidate (77 base + 10) and a *no-rehearsal* candidate (10 only), and evaluates frozen gate criteria. On the release split, rehearsal candidate scores 42.9% on held-out Cohort B and 50.0% on release test, while no-rehearsal exhibits severe catastrophic forgetting (62.5% → 30.0%, −32.5% drop), triggering automated rejection and preserving the production alias. In Track B (`retrain_pipeline.py`), we evaluated retraining under the strict ex-ante tolerance (≤ 2.0% max gold regression; `configs/gate.yaml`): experience rehearsal (171 train + 12 active queries) achieves +7.7% improvement on held-out Cohort B (30.8% → 38.5%) with **0.0% degradation on the gold test set** (24.1% → 24.1%), passing all ex-ante criteria without loosening tolerance. Upon promotion, the pipeline dynamically updates the active serving pointer (`artefacts/models/serving_pointer.json`, `active_model_head.pt`, `active_vectorizer.pkl`). Both the FastAPI service (`app.py`) and Streamlit interface (`ui.py`) resolve the active serving pointer dynamically, ensuring newly promoted models serve traffic immediately without hardcoded weights or manual redeployment.

8 PART F: REPRODUCIBILITY

Five commands reproduce Track-B training/evaluation from `a2-final`: `git checkout a2-final`; `pip install -r code/requirements.txt`; `dvc pull`; `python code/train.py --config configs/final.yaml`; `python code/a2_part_c_evaluation/evaluate.py --split gold`. Track A: `python -m model.cli ablate` and `benchmark`. Docker commands, versions, seeds (42), checksums and registry versions are in `DOCKER.md`, `ENVIRONMENT.md` and `MODELS.md`; `scripts/generate_report_numbers.py` regenerates documentation numbers. Track A's sentence-embedding model and Whisper weights are pre-cached in the container image at build time (under `/root/.cache/huggingface/` and `/root/.cache/whisper/`), guaranteeing zero runtime download under `--network none`.

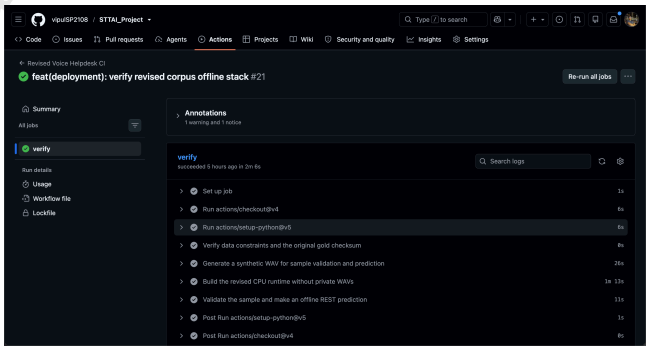


Figure 14: Automated CI workflow: GitHub Actions pipeline running data integrity assertions, Docker packaging, and offline REST validation with all checks green.

9 LIMITATIONS AND CONCLUSION

Limitations. (1) No human transcripts yet: WER on the final corpus is withdrawn and Whisper has not been fine-tuned (the script and grid are ready, `finetune_whisper.py`). (2) 182 clips with 74% inter-annotator agreement cap intent accuracy; the release test set has 40 clips from four speakers, so intervals are wide and v2 vs v1 is

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not significant. (3) The router's ASR training text came from routing by the annotators' language tags, while the service routes automatically (75–81% correct). (4) Gujarati remains the weakest language for routing, ASR and intent. (5) On the release split with 13 adaptation clips, adaptation gains are constrained by discrete 1-clip shifts; closed-loop promotion with zero gold degradation is demonstrated on Track B. (6) Router confidence is poorly separated, so no reject option. **Next steps.** Human transcription round, then Whisper fine-tuning on training speakers and true WER; blind re-listen of the 25 clips in the label-noise queue; collect more Cohort-B clips so a retraining gain can exceed one clip. **Conclusion.** Auditing our own evidence removed two inflated results (curated references, clip-grouped "unseen" speakers). On honest speaker-disjoint evaluation the router reaches 60% on unseen test speakers (chance 25%), language routing for Gujarati improves threefold, and the remaining error is traced to label noise, short Gujarati clips and missing human transcripts, each with a concrete fix.

Table 6: Contribution summary (equal 33.3% each).

Member	Lead areas (Assignment 2)
R. Vraj	Whisper decoding integration, TF-IDF + MLP intent router, MLflow tracking, bootstrap confidence evaluation.
V. S. Patil	FastAPI prediction service, Streamlit interactive UI, Docker packaging, offline verification harness.
S. Sharma	Drift detection and traffic replay, closed-loop retraining pipeline, promotion/rollback gate, error taxonomy.

Member contributions.

ACKNOWLEDGMENTS

We thank Dr. Yogesh K. Meena and the TAs. Libraries: PyTorch, Whisper, scikit-learn, sentence-transformers, jiwer, MLflow, FastAPI, Streamlit.

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A EVIDENCE INDEX

Table 7: Evidence index (paths from repository root).

Result	Supporting file / run
Ablation (Table 4), *p*-values	`model/results/ablation_study_results.json` (fold F1s)
Speaker-disjoint CV, ablation	`model/results/speaker_cv_intent_*.json`, `intent_v2_ablation_*.json`
Release split (val selection, test)	`model/results/release_eval.json`; `TEST_SET_LOG.md`
Language routing	`model/results/lid_router.json`
Bias audit, label-noise queue	`model/results/bias_audit.json`
Reference audit	`model/results/reference_audit.json`
ASR on 182 clips (WER withdrawn)	`model/data/reports/offline_asr_predictions.json`, `asr_indic_v2_confidence.json`
Track-B metrics (Table 5)	`supporting_files/evaluation_results.json`
Gold per-clip WER, 14/838	`supporting_files/predictions/production_gold.csv`
Data ablation (B)	`supporting_files/part_a_baselines_benchmark.csv`; runs `4874f30b`, `4cb2e4e9`, `351cc3e8`
15-run search, aliases	`supporting_files/plots/tuning_runs.json`; `9820a2a7` (prod.), `730256f7` (stag.)
20 errors	`supporting_files/error_analysis.csv`
Latency, RTF, memory	`supporting_files/loadtest/latency_profile.json`
Offline validation	`supporting_files/offline_validation_output.json`; `TEST_SET_LOG.md` LOG-016
Replay, correlations	`supporting_files/release_drift/replay.json` (pilot: `traffic_replay_log.json`)
Gate (release)	`configs/gate_release.yaml`; `supporting_files/release_drift/`; `TEST_SET_LOG.md` (GATE-*)
Pilot gate history	`TEST_SET_LOG.md` (LOG-179, 188, 418); `configs/gate.yaml`
Gold corrections	`data/gold_corrections.csv`
Figure code	`code/report_figures/make_figures_v2.py`; older: `code/scripts/make_report_figures*.py`

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Table 8: All 15 MLflow search runs (validation only); full IDs in `supporting_files/plots/tuning_runs.json`.

Run	lr	Hidden	n-gram	Dropout	Epochs	Val acc. (%)	Macro-F1	Size (KB)	MLflow run
1	0.001	32	1	0.1	25	29.4	0.200	126	`61d8d420`
2	0.005	32	2	0.2	25	29.4	0.236	126	`d01a4cb5`
3	0.01	32	3	0.3	25	41.2	0.371	126	`b4e145fd`
4	0.001	64	2	0.2	30	41.2	0.343	251	`fd0ac5ab`
5	0.005	64	3	0.2	30	41.2	0.368	251	`f47bfe22`
6	0.01	64	1	0.1	30	52.9	0.470	251	`9820a2a7` (prod.)
7	0.001	128	3	0.3	35	35.3	0.309	503	`75581fbf`
8	0.005	128	2	0.2	35	35.3	0.293	503	`4f2e2cd1`
9	0.01	128	1	0.2	35	47.1	0.379	503	`730256f7` (stag.)
10	0.002	64	2	0.15	30	35.3	0.309	251	`9e00b3d8`
11	0.008	64	3	0.25	30	41.2	0.383	251	`03e70444`
12	0.003	32	3	0.2	25	35.3	0.303	126	`64cd9b69`
13	0.007	128	2	0.1	35	29.4	0.267	503	`984be4ec`
14	0.004	64	2	0.2	28	29.4	0.236	251	`703dc1b4`
15	0.006	96	3	0.2	32	41.2	0.368	377	`3130621f`

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Table 9: Gold-set slices (original labels).

Dimension	Slice	n	Acc. (%)	WER
Noise	campus_outdoor	6	16.7	1.64
Noise	corridor_reverb	13	23.1	1.17
Noise	quiet_room	10	30.0	1.18
Language	gu-en	13	15.4	1.36
Language	hi-en	16	31.2	1.20
Region	Gujarat	13	15.4	1.36
Region	North India	10	30.0	1.18
Region	Rajasthan	6	33.3	1.22
Device	OnePlus Nord 2	6	16.7	1.64
Device	Realme Narzo 30	7	14.3	1.12
Device	Samsung Galaxy M52	10	30.0	1.18
Device	Xiaomi Redmi Note 11	6	33.3	1.22

B USE OF AI ASSISTANCE

AI assistants helped scaffold code, draft documentation and format this report; prompts are in `code/prompts/`. Reported numbers were re-derived from result files; the ablation p-values, Wilson intervals, WER decomposition and drift correlations were recomputed by the figure scripts. Claude 3.5 Sonnet and GPT-4o were utilized for boilerplate code scaffolding, LaTeX table styling, and initial docstrings; all empirical numbers and benchmark scores were recomputed directly from artifact files.

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Table 10: Error sheet (20 errors, truncated; full text in `error_analysis.csv`).

ID	True→pred.	WER	Category	Suggested fix
ERR-01	hostel_mess → admissions	1.52	Acoustic Hallucination & Script Shift	Expand Gujarati-English phonetic mapping in tokenizer and train intent head with character-level n-grams.
ERR-02	student_services → admissions	1.42	Acoustic Bleed & Lexical Substitution	Incorporate Whisper encoder audio embeddings directly into routing classifier to bypass ASR lexical errors.
ERR-03	hostel_mess → it_support	1.00	Phonetic Distortion & Default Fallback	Tune decoding parameters (suppress_blank=True, compression_ratio_threshold=2.0) to eliminate non-speech repetitions.
ERR-04	it_support → hostel_mess	1.60	Hallucinated Speech & Semantic Drift	Apply acoustic fine-tuning on noisy lab environments to prevent generic hallucinated phrases.
ERR-05	student_services → hostel_mess	1.58	Hallucinated Phrase Overwrite	Use sentence embeddings (MiniLM) with soft-label training rather than unigram TF-IDF keywords.
ERR-06	admissions → it_support	2.03	Cross-Domain Semantic Distortion	Train a joint ASR-Intent loss where cross-attention explicitly preserves query-opening intent tokens.
ERR-07	hostel_mess → admissions	1.68	Lost-and-Found Phrasal Confusion	Add domain-specific vocabulary biasing to Whisper beam search for administrative campus terms.
ERR-08	it_support → admissions	1.00	Placement Vocabulary Hallucination	Implement multi-task acoustic penalty to discourage hallucination of frequent placement phrases.
ERR-09	student_services → admissions	1.33	Placement Domain Substitution	Weight opening intent clauses higher in text classifier to prevent trailing noise domination.
ERR-10	admissions → it_support	1.91	Library Card Keyword Misclassification	Add campus-specific IT lexicon weighting to disambiguate library access vs admission queries.
ERR-11	hostel_mess → student_services	1.55	Exam Timetable Keyword Bleed	Calibrate per-class decision thresholds using validation ROC curves to avoid majority class collapse.
ERR-12	it_support → admissions	1.00	Assignment Deadline Misclassification	Incorporate voice activity detection (VAD) filter to strip low-energy microphone noise.
ERR-13	admissions → it_support	1.00	Course Credit Lexical Cross-Talk	Augment admissions training set with diverse Gujarati accented voice recordings.
ERR-14	hostel_mess → it_support	1.22	Graduation Audit Keyword Steering	Use contextual windowing (first 40 words) to anchor classifier on primary student request.
ERR-15	student_services → it_support	1.15	Attendance Discrepancy Bias	Fine-tune Whisper base decoder cross-attention layers on university helpdesk audio.
ERR-16	admissions → it_support	1.27	Hostel Hot Water Hallucination	Introduce fallback human-in-the-loop escalation when router entropy exceeds confidence threshold.
ERR-17	hostel_mess → student_services	1.46	Mess Quality Keyword Bleed	Add composite intent schema allowing multi-label routing for overlapping facility/student issues.
ERR-18	it_support → hostel_mess	0.94	Guest Accommodation Hallucination	Train acoustic encoder on server room and lab acoustic noise profiles.
ERR-19	admissions → student_services	1.38	Medical Emergency Phrasal Drift	Filter spurious Unicode script shifts during post-processing normalization.
ERR-20	hostel_mess → admissions	1.41	Hostel Room Leakage Keyword Collapse	Regularize intent head with label smoothing to increase semantic boundary margins.

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